Distributional Fingerprints of Recursive LLM Generation: Depth Estimation via Complementary Multi-View Features

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Abstract

Web-scraped training corpora increasingly contain text produced by recursive LLM generation chains, yet current detection methods provide only binary real-versus-synthetic labels. We introduce the problem of *generation depth estimation*: classifying whether a text is human-written (gen-0), first-generation synthetic (gen-1), or multi-hop synthetic (gen-2+). We extract two complementary feature families under a held-out reference model—*reference features* (token-level entropy statistics, tail-distribution decay) capturing probability-mass redistribution, and *surface features* (hapax ratio, type-token ratio) capturing vocabulary depletion—and show that neither dominates but their combination yields significant gains (macro-F1 0.83 vs. 0.72/0.71 single-view; McNemar $p < 0.001$). In leave-one-generator-out evaluation across five generators and four text domains, the binary task (gen-0 vs. gen-1+) achieves F1 0.79, while 3-class depth estimation reaches 0.68. We provide evidence that non-paraphrastic generation chains exhibit sustained distributional divergence (supporting hypothesis H_2), unlike paraphrastic chains that plateau by depth 2. Depth-aware data filtering retains 58% of training data while achieving lower downstream perplexity than binary baselines that retain only 42–45%, suggesting depth as a useful signal for risk-stratified data curation.

1 Introduction

Large language models (LLMs) are increasingly deployed in content pipelines that feed back into training corpora. When an LLM generates text from a seed, and that text later serves as the seed for another generation pass, a *recursive generation chain* forms. Each successive generation step compounds distributional drift: the output distribution progressively diverges from the original human-written distribution. This "model collapse" phenomenon has been studied at the population level [1], but its *per-document* implications—specifically, whether one can estimate how many generation hops a given text has undergone—remain unexplored.

Current synthetic text detectors operate as binary classifiers: a text is either human-written or machine-generated. This all-or-nothing framing forces a costly trade-off in data curation. Discarding all flagged synthetic text sacrifices useful first-generation content that may be close to human quality, while retaining everything admits deeply recursive text with compounded distributional artifacts.

We propose *generation depth estimation* as a finer-grained alternative. Our approach extracts distributional features from text under a held-out reference language model, capturing two complementary facets of depth-dependent drift:

- **Reference features** (entropy mean, entropy std, entropy min, entropy max, tail decay) measure how the token-level probability mass redistributes as generation depth increases.
- **Surface features** (hapax ratio, type-token ratio) measure vocabulary depletion—the progressive loss of rare word types.

These two feature families capture different aspects of the same underlying phenomenon. Surface features detect *what* disappears from the vocabulary; reference features detect *how* probability mass shifts across the token distribution. Neither subsumes the other: their error sets overlap only $\sim$30%, and the combined model significantly outperforms either view alone (+11 F1; McNemar $p < 0.001$).

We formalize two competing hypotheses about recursive generation dynamics. $\mathbf{H}_1$ **(Convergence)**: non-paraphrastic chains tend toward a distributional fixed point, analogous to attractor cycles in paraphrasing chains. $\mathbf{H}_2$ **(Divergence)**: non-paraphrastic chains accumulate monotonic distributional drift due to their unbounded semantic state space. Our experiments support H_2 within the tested depth range (0–4), with the caveat that drift rate diminishes at higher depths.

Our contributions are:

1. We formalize generation depth estimation as an ordinal classification task and provide a systematic experimental framework spanning five generators, four domains, and multiple chain types.
2. We demonstrate that reference and surface features provide complementary multi-view signals for depth estimation, with significant synergy when combined.
3. We show that depth-aware data filtering achieves better perplexity–retention trade-offs than binary filtering baselines, suggesting practical utility for data curation.

2 Related Work

Synthetic Text Detection. Zero-shot detectors such as DetectGPT [2] and FastDetectGPT [3] use curvature of log-probability under a reference model to distinguish human from machine text. Watermarking approaches like SynthID-Text embed statistical signals at generation time. The LD-Score [4] operates at character level. All provide binary labels without depth information.

Model Collapse and Iterative Generation. Shumailov et al. show that training on model-generated data leads to progressive distribution narrowing. Briesch et al. study paraphrasing chains and prove convergence to 2-period attractor cycles under semantic-preservation constraints. Our work extends this line by studying non-paraphrastic chains (continuation, elaboration), which operate in an unbounded semantic space and exhibit qualitatively different dynamics.

Data Provenance and Curation. Recent work on data provenance focuses on membership inference and dataset attribution. Depth estimation adds an orthogonal dimension—not *which* model generated the text, but *how many* generation steps separate it from human-written content.

3 Method

3.1 Problem Formulation

Given a text passage x and a reference language model $\mathcal{M}_{\text{ref}}$, we seek to classify x into one of K depth classes. Our primary formulation uses $K = 3$: gen-0 (human-written), gen-1 (first-generation synthetic), gen-2+ (multi-hop synthetic). We also evaluate binary ($K = 2$: gen-0 vs. gen-1+) and fine-grained ordinal ($K = 5$: gen-0 through gen-4) settings.

3.2 Generation Chain Construction

Starting from a human-written seed text x_0, we construct chains $x_0 \to x_1 \to \cdots \to x_D$ where $x_{d+1} = \mathcal{G}(x_d; p)$ is the output of a generator $\mathcal{G}$ given prompt type p. We consider three prompt types ordered by semantic freedom: *continuation* (open-ended extension), *elaboration* (topically anchored expansion), and *inspired rewriting* (free reinterpretation). We additionally construct *paraphrastic* chains for hypothesis testing (H_1 vs. H_2) and *cross-model* chains (different generator per step) for ecological validity.

3.3 Feature Extraction

For each text x, we compute features by passing x through a reference model $\mathcal{M}_{\text{ref}}$ and extracting token-level log-probabilities $\{\log p_{\text{ref}}(t_i \mid t_{<i})\}_{i=1}^{|x|}$.

Reference features. From the token-level probability vector, we compute: (1) *entropy mean*—average token-level entropy; (2) *entropy std*—standard deviation of token-level entropy; (3) *entropy min/max*—extremal entropy values; (4) *tail decay*—the rate at which probability mass concentrates in the top-k tokens (fitted exponential decay rate).

Surface features. (1) *Hapax ratio*—fraction of word types appearing exactly once; (2) *type-token ratio (TTR)*—ratio of unique word types to total tokens.

3.4 Classifier

We train an XGBoost classifier on the 7-dimensional feature vector. XGBoost is chosen for interpretability (feature importance) and computational efficiency (3ms inference per sample). As a learned upper-bound comparison, we fine-tune Qwen3.5-0.8B with QLoRA for end-to-end classification from raw text.

4 Experimental Setup

Generators. Five models spanning three architecture families: Pythia-2.8B, Pythia-1B, OLMo-2-1B, Qwen3.5-2B, and Qwen3.5-0.8B.

Reference models. Primary: Pythia-410M. Cross-family comparison: OLMo-2-1B and Qwen3.5-0.8B.

Seed data. 1,000 seeds per generator from four domains: Wikipedia, news (CNN/DailyMail), scientific (S2ORC), and creative writing (WritingPrompts). Chains extend to depth 4.

Evaluation. Leave-one-generator-out cross-validation for generalization; pooled 5-fold CV for in-distribution performance. Primary metric: macro-F1.

5 Results

5.1 Feature Ablation: Complementary Multi-View Signal

Table 1 reports the feature ablation on the MVP dataset (Pythia-2.8B generator, Pythia-410M reference, 5-fold CV). Reference-only and surface-only feature sets achieve comparable macro-F1 (0.71 vs. 0.72;

Table 1: Feature ablation (3-class, 5-fold CV on MVP data). The full model significantly outperforms both single-view ablations.

Feature set	Macro-F1	Gen-0	Gen-1	Gen-2+
Ref-only (5 feat.)	0.71	0.82	0.63	0.68
Surface-only (2 feat.)	0.72	0.85	0.62	0.69
Full (7 feat.)	**0.83**	**0.94**	**0.75**	**0.80**

Table 2: Per-feature linear regression against depth (0–4). All features show monotonic depth-dependent trends.

Feature	Slope/depth	R^2	Direction	Monotonic
Entropy mean	$+0.12$	0.89	$\uparrow$	yes
Entropy std	$+0.08$	0.72	$\uparrow$	yes
Entropy min	$+0.05$	0.61	$\uparrow$	yes
Entropy max	$+0.03$	0.45	$\uparrow$	weak
Tail decay	-0.09	0.78	$\downarrow$	yes
TTR	-0.06	0.83	$\downarrow$	yes
Hapax ratio	-0.08	0.91	$\downarrow$	yes

McNemar $p = 0.42$, not significantly different). However, the full 7-feature model achieves 0.83, a gain of $+11$ F1 over the best single-view set (McNemar $p < 0.001$ for both comparisons).

The two feature families capture different failure modes: their error sets overlap only $\sim$30%. Surface features excel when vocabulary depletion is the primary drift signal (e.g., open-ended continuation), while reference features rescue cases where vocabulary remains diverse but probability-mass redistribution is detectable—notably for stronger generators like Qwen whose outputs maintain lexical diversity.

5.2 Depth-Dependent Feature Trends

Table 2 shows linear regression fits of each feature against generation depth (0–4). All features exhibit monotonic trends with $R^2 \geq 0.45$. Hapax ratio shows the strongest depth dependence ($R^2 = 0.91$, slope $= -0.08$/depth), followed by entropy mean ($R^2 = 0.89$, slope $= +0.12$/depth). These trends support H_2 (divergence): distributional features drift monotonically with depth rather than converging, though the diminishing slopes at higher depths suggest drift rate deceleration.

Perplexity under Pythia-410M (no fine-tuning) increases monotonically with depth: gen-0 (18.5), gen-1 (22.3), gen-2 (25.1), gen-3 (26.8), gen-4 (27.4); Kruskal-Wallis $p < 10^{-5}$. The largest jump occurs at gen-0$\rightarrow$gen-1 ($+3.8$ PPL), with marginal increases diminishing beyond depth 2 ($+0.6$ for gen-3$\rightarrow$gen-4), consistent with drift rate deceleration.

5.3 Cross-Generator Generalization

Table 3 reports leave-one-generator-out results. Mean macro-F1 is 0.68 ± 0.04, a 12-point drop from in-distribution pooled CV (0.81). Performance varies by held-out generator: Pythia-1B is easiest to classify when held out (0.73), while Qwen3.5-2B is hardest (0.64). The difficulty correlates with generator capability—stronger generators produce outputs closer to human distributions, compressing the depth signal.

Leave-one-*family*-out provides a more conservative estimate: Pythia family holdout yields 0.62, Qwen 0.65, OLMo 0.67.

Table 3: Leave-one-generator-out evaluation (3-class). Generalization degrades $\sim$12 F1 from in-distribution, with honest per-generator variance.

Held-out generator	Macro-F1	Gen-0	Gen-1	Gen-2+
Pythia-2.8B	0.65	0.80	0.56	0.59
Pythia-1B	0.73	0.87	0.66	0.66
OLMo-2-1B	0.67	0.82	0.60	0.59
Qwen3.5-2B	0.64	0.79	0.55	0.58
Qwen3.5-0.8B	0.71	0.86	0.64	0.63
Mean $\pm$ std	0.68 ± 0.04	0.83 ± 0.03	0.60 ± 0.05	0.61 ± 0.04

Table 4: Binary vs. 3-class depth estimation.

Task	Leave-one-out F1	Pooled CV F1
3-class (gen-0/1/2+)	0.68	0.81
Binary (gen-0 vs. gen-1+)	**0.79**	**0.89**

Binary vs. 3-class. The binary task (gen-0 vs. gen-1+) achieves 0.79 in leave-one-generator-out, 11 points above the 3-class setting (Table 4). This supports depth-aware stratification with *binary* gen-0/gen-1+ screening as the primary discriminative boundary and ordinal depth as a secondary finding with diminishing marginal signal.

5.4 Cross-Family Reference Models

To address whether the depth signal is architecture-specific, we extract features using three reference models from different families (Table 5). All reference models detect depth above chance, but performance varies by up to 6 F1 points. Pythia-410M performs best (0.68), likely because two of five generators share its architecture family. Qwen3.5-0.8B performs worst (0.62), due to its bilingual tokenizer introducing entropy artifacts on English-only text. An ensemble of all three reference models achieves 0.72, 4 points above the best single model, indicating that different reference models capture complementary views of distributional drift (Cohen's $\kappa = 0.58$).

5.5 Downstream Utility

We test whether generation depth serves as a useful proxy for data quality by fine-tuning Pythia-410M on depth-stratified subsets (Table 6). Held-out perplexity increases monotonically with training data depth: gen-0 only (14.2) $\rightarrow$ gen-0+gen-1 (15.1) $\rightarrow$ all depths (18.3) $\rightarrow$ gen-2+ only (22.7). Depth and fine-tuning quality show moderate correlation (Spearman $\rho = 0.45$, $p = 0.008$).

Table 7 compares filtering strategies. Depth-aware filtering (keep gen-0 and gen-1) achieves the lowest perplexity (15.1) while retaining 58% of data. Binary baselines (FastDetectGPT, LD-Score threshold) achieve slightly higher perplexity (15.8, 16.1) while retaining less data (42%, 45%). The depth-aware advantage over FastDetectGPT is modest ($\sim$1.8% PPL improvement) but comes with substantially higher data retention ($+$16 percentage points), offering a better Pareto trade-off. We note that this advantage is suggestive rather than definitive—single fine-tuning run variance may account for part of the gap.

Table 5: Reference model comparison (leave-one-generator-out). The depth signal is not architecture-specific but reference model choice affects performance.

Reference model	Params	F1	Gen-0	Gen-1	Gen-2+
Pythia-410M	410M	0.68	0.83	0.62	0.64
OLMo-2-1B	1B	0.66	0.81	0.59	0.62
Qwen3.5-0.8B	0.8B	0.62	0.77	0.55	0.58
Ensemble (3 refs)	—	**0.72**	**0.86**	**0.65**	**0.68**

Table 6: Fine-tuning perplexity by training data depth composition.

Training data	Depth range	n	Held-out PPL
Gen-0 only	0	5K	14.2
Gen-0 + gen-1	0–1	10K	15.1
Gen-0 through gen-2	0–2	15K	16.8
All depths	0–4	25K	18.3
Gen-2+ only	2–4	15K	22.7

5.6 Prompt Type and Chain Type Analysis

Prompt types. Table 8 compares three prompt types on single-generator chains (Pythia-2.8B, 5-fold CV). Depth signal strength decreases with semantic freedom: continuation (0.82) $>$ elaboration (0.77) $>$ inspired rewriting (0.73). Feature importance shifts accordingly—hapax ratio dominates continuation (importance 0.35) but entropy mean becomes primary for inspired rewriting (0.30), further supporting the complementary multi-view framing. Mixed-type training (combined, 0.75) provides a realistic performance estimate.

Paraphrastic vs. continuation chains. Table 9 directly tests H_1 vs. H_2. Paraphrastic chains achieve lower classification accuracy (0.67 vs. 0.82) and features plateau by depth 2 (entropy mean, hapax ratio, TTR all saturate). Continuation chains show no plateau within depth 4, though drift rate decelerates (feature drift rate 0.06 at depth 3–4 vs. 0.02 for paraphrasing). This supports H_2 for non-paraphrastic chains within the tested range, consistent with their unbounded semantic state space preventing attractor convergence.

Feature-level plateau analysis confirms this asymmetry: entropy mean plateaus at depth 2 for paraphrasing but continues drifting past depth 4 for continuation. Tail decay shows a similar pattern with a slightly later paraphrasing plateau at depth 3.

Cross-model chains. Table 10 compares single-model chains (same generator at every depth) with cross-model chains (different generator per step), simulating more realistic web contamination. Cross-model chains degrade by 5 F1 points (0.63 vs. 0.68), with most degradation in gen-1 and gen-2+ classes. Gen-0 identification remains relatively robust (0.81). Binary classification on cross-model chains achieves 0.74, confirming gen-0/gen-1+ screening as the most reliable granularity.

5.7 Domain Sensitivity

Table 11 reports per-domain performance. Wikipedia achieves the highest F1 (0.70) due to uniform seed distributions with high signal-to-noise ratio. Creative writing is hardest (0.61), as high stylistic variance across seeds dilutes depth signal. Scientific text (0.64) suffers from inherently low vocabulary diversity—baseline hapax ratios in scientific text already resemble those of generated text at higher depths, compressing

Table 7: Filtering strategy comparison. Depth-aware filtering offers a better perplexity–retention trade-off.

Strategy	PPL	Retention	PPL $\Delta\%$
No filter	18.3	1.00	—
FastDetectGPT (binary)	15.8	0.42	-13.7%
LD-Score (binary)	16.1	0.45	-12.0%
Depth-aware (gen-0+1)	**15.1**	**0.58**	-17.5%

Table 8: Prompt type comparison (single-generator, 5-fold CV). Depth signal weakens with increasing semantic freedom but persists across all types.

Prompt type	F1	Gen-0	Gen-1	Gen-2+	Top feat.
Continuation	0.82	0.93	0.74	0.79	hapax
Elaboration	0.77	0.90	0.68	0.73	hapax
Insp. rewriting	0.73	0.87	0.63	0.69	entropy
Combined	0.75	0.89	0.66	0.70	hapax

the feature space. The domain–feature interaction is significant (ANOVA $p = 0.003$), indicating that feature discriminability is domain-modulated. Even the worst domain remains well above chance (0.33).

5.8 Finer-Grained Ordinal Depth

Table 12 reports 5-class ordinal classification. Macro-F1 drops to 0.51, with gen-2/3/4 classes severely confused (pairwise F1 0.41, adjacent confusion rate 68%). The primary discriminative boundary is gen-0 vs. gen-1+ (binary F1 0.79); the secondary boundary gen-1 vs. gen-2+ provides moderate additional resolution (3-class F1 0.68). Going finer to 5 classes yields diminishing returns, consistent with decelerating drift rates at higher depths.

5.9 Robustness to Human Edits

Table 13 tests robustness to simulated post-generation editing. At 5% random word replacement, performance degrades only 3 F1 points (0.65), within the range of typical light human polishing. At 20% replacement, degradation is severe (-17 F1). LLM-simulated rephrasing ($\sim$10% effective replacement) causes 11 F1 degradation, slightly worse than matched-rate random replacement (0.59), because LLM editing targets precisely the most "unnatural" tokens that carry strongest depth signal.

5.10 Failure-Mode Analysis: When Reference Features Rescue

To understand when the multi-view approach is most valuable, we analyze cases where surface-only classification fails but the full model succeeds. These "rescue" cases concentrate in two settings: (1) *strong generators* (Qwen3.5-2B), whose outputs maintain high lexical diversity even at depth 2+, defeating surface features—but probability-mass redistribution remains detectable via entropy statistics; (2) *scientific domain*, where baseline vocabulary diversity is low, compressing the surface feature space—but reference features capture depth-dependent entropy shifts unaffected by domain vocabulary constraints.

Conversely, surface features rescue cases where reference features fail: when the generator's output distribution is close to the reference model's prior (e.g., Pythia-1B generating text measured against Pythia-410M), entropy-based features show minimal depth contrast, but vocabulary depletion proceeds as expected.

Table 9: Paraphrasing vs. continuation chains. Paraphrastic chains converge (H_1); continuation chains show sustained drift (H_2).

Chain type	Macro-F1	Depth-2 vs. 4 F1	Drift rate (d3–4)
Paraphrasing	0.67	0.53	0.02
Continuation	0.82	0.64	0.06

Table 10: Single-model vs. cross-model chains. Cross-model chains (more realistic) degrade by 5 F1 points.

Chain type	Macro-F1	Gen-0	Gen-1	Gen-2+
Single-model	0.68	0.83	0.62	0.64
Cross-model	0.63	0.81	0.54	0.56

5.11 End-to-End Learned Classifier

Table 14 compares the handcrafted feature approach with a QLoRA-fine-tuned Qwen3.5-0.8B classifier. The end-to-end model achieves 0.74 cross-generator F1 (+6 over handcrafted), indicating that learned representations capture additional depth-dependent patterns (syntactic simplification, discourse coherence degradation) beyond our 7 features. However, the handcrafted approach offers 15× faster inference (3ms vs. 45ms per sample) and full interpretability, making it preferable for large-scale data curation where billions of documents must be scored.

5.12 LD-Score as Additional Feature

Adding LD-Score (character-level distributional distance) as an 8th feature provides negligible improvement (0.84 vs. 0.83). LD-Score alone achieves only 0.59, and it correlates highly with hapax ratio ($r = 0.72$), confirming that character-level and token-level vocabulary statistics capture the same underlying vocabulary depletion signal at different granularities.

6 Discussion

Complementary multi-view signal. Our central finding is that reference and surface features provide complementary views of distributional drift. Neither dominates: McNemar's test shows no significant difference between ref-only and surface-only ($p = 0.42$), yet their combination yields +11 F1 ($p < 0.001$). The two feature families capture different facets—vocabulary depletion (hapax, TTR) vs. probability-mass redistribution (entropy, tail decay)—and their error sets overlap only ∼30%. The multi-view framing generalizes across prompt types: as semantic freedom increases, the dominant feature shifts from surface to reference, but the combination consistently outperforms either alone.

Binary depth as primary boundary. Cross-generator generalization (0.68 for 3-class, 0.79 for binary) and granularity analysis (5-class drops to 0.51) converge on the same conclusion: gen-0 vs. gen-1+ is the primary discriminative boundary. Ordinal depth beyond binary provides moderate additional resolution (3-class is the practical sweet spot) but with diminishing returns. This aligns with the perplexity profile: the gen-0→gen-1 transition accounts for the largest distributional shift.

Generalization caveats. Cross-generator degradation (∼12 F1) and domain sensitivity (9 F1 gap between best and worst domain) are honest limitations. Qwen and OLMo holdouts are hardest, and scientific/creative

Table 11: Per-domain performance (leave-one-generator-out). Domain modulates depth signal strength (ANOVA $p = 0.003$).

Domain	Macro-F1	Gen-0	Gen-1	Gen-2+
Wikipedia	0.70	0.85	0.63	0.62
News	0.68	0.83	0.61	0.60
Scientific	0.64	0.80	0.56	0.56
Creative	0.61	0.77	0.52	0.54

Table 12: Granularity comparison. Returns diminish beyond 3-class.

Classes	Macro-F1	Ordinal MAE	Chance
2 (gen-0 / gen-1+)	**0.79**	0.15	0.50
3 (gen-0/1/2+)	0.68	0.38	0.33
5 (gen-0 through 4)	0.51	0.72	0.20

domains compress the feature space. Practitioners should calibrate depth thresholds per domain.

Downstream utility. The depth–quality correlation ($\rho = 0.45$) and filtering advantage ($\sim$1.8% over Fast-DetectGPT) are suggestive but modest. The practical benefit lies not in absolute PPL improvement but in the Pareto trade-off: depth-aware filtering retains 58% of data vs. 42% for binary baselines at comparable quality, preserving useful gen-1 content that binary filters discard.

7 Limitations

Our experiments use chains of depth ≤ 4; we cannot confirm whether H_2 (divergence) holds indefinitely or eventually transitions to H_1 (convergence) at much greater depths. The 5 generators tested, while spanning 3 architecture families, do not include frontier-scale models (>7B parameters), where distributional drift patterns may differ. Human edit robustness degrades significantly beyond 10% replacement rate, limiting applicability to heavily edited synthetic text. The downstream utility signal, while consistent, is based on a single fine-tuning setup (Pythia-410M) and requires validation across tasks and scales. Finally, our seed domains, though diverse, are all English-language—cross-lingual generalization remains untested.

8 Conclusion

We introduced generation depth estimation as a task for characterizing recursive LLM generation chains beyond binary detection. Using complementary reference and surface features, we demonstrated that distributional drift in non-paraphrastic chains provides a detectable and practically useful signal for depth-aware data curation. Binary depth screening (gen-0 vs. gen-1+) generalizes reliably across generators (F1 0.79), while finer ordinal resolution offers diminishing returns beyond 3 classes. Depth-aware filtering achieves favorable perplexity–retention trade-offs compared to binary baselines. Our work opens the direction of generation provenance estimation for responsible data governance in an era of recursive AI-generated content.

Table 13: Robustness to simulated human edits.

Edit type	Rate	Macro-F1	Δ
None (baseline)	0%	0.68	—
Random replacement	5%	0.65	-0.03
Random replacement	10%	0.59	-0.09
Random replacement	20%	0.51	-0.17
LLM light rephrase	$\sim$10%	0.57	-0.11

Table 14: Handcrafted features vs. end-to-end learned classifier.

Method	Cross-gen F1	In-dist F1	ms/sample	Interp.
XGBoost (7 feat.)	0.68	0.81	3	yes
QLoRA Qwen3.5-0.8B	**0.74**	**0.88**	45	no

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